Chapter 1 Holomorpihc Functions in Several Variables

1.1 Fundamentals

1.1.1 Analysis and Topologies

Recall

- Power Series $\mathbb{C}[[z_1, \dots, z_n]]$ with elements $\sum a_{i_1,\dots,i_n} z_1^{i_1} \dots z_n^{i_n}$, $a_{i_1,\dots,i_n} \in \mathbb{C}$;
- Convergence of Power Series;
- Uniformly Convergence to function $f : \mathbb{C}^n \supseteq M \to \mathbb{C}$,

$$|\sum a_{i_1,\dots,i_n} z_1^{i_1} \dots z_n^{i_n} - f(z)| < \epsilon, \forall z \in M$$

- Holomorphic Function $\Leftrightarrow$ Analytic

  According to Cauchy's theorem in multivariables, we obtain the equivalence.

$$f \text{ is holomorphic} \Leftarrow f \text{ is analytic:} \forall z_0 \in M, \exists U(z_0), \sum a_v (z - z_0)^v \text{converges on } U(z_0) \text{ to } f(z)$$

- Absolute Space: A pointed set

$$V = \{(z_1, \dots, z_n) \in \mathbb{C}^n : |z_i| < r_i\}$$

  With the modular mapping on each component,

$$\tau : \mathbb{C}^n \to V \qquad (z_1, \dots, z_n) \mapsto (|z_1|, \dots, |z_n|)$$

Polycylinder

Definition 1.1.1.1 Let $\tau \in V$ be an element in absolute space, $z_0 \in \mathbb{C}^n$.
then define the **polycylinder about** z_0 **with radius** τ as

$$\Delta(z_0, \tau) = \{z \in \mathbb{C}^n : |z_i - z_{0,i}| < \tau_i, i = 1, \dots, n\}$$

And, $\partial\Delta(z_0, \tau)$ is the **distinguished boundary** of $\Delta(z_0, \tau)$.

It is easy to see that the geometric shape of polycylinder is actually a **product of discs** in $\mathbb{C}^n$. Thus, the boundary is a product of circles, actually 'torus' in $\mathbb{C}^n$.

Now, take the following notion

$$\odot\mathbb{C}^n :=' CC^n - \{0\}^n \qquad P_\tau = \Delta(0, \tau)$$

1.1.2 Rheinhardt Domain

Torus Action

Definition of Rheinhardt Domain

Definition 1.1.2.1 Let $D \subseteq \mathbb{C}^n$ be a domain, then D is called a **Rheinhardt Domain** if for any $z \in D$, the polycylinder $\Delta(0, |z|) \subseteq D$.

1.1.3 Complex Differential and Cauchy-Riemann Equation

Definition 1.1.3.1 ...

1.1.4 Some of Conclusions

◇ **Lemma 1.1.4.1** ...

1.2 Theorems

1.2.1 Cauchy's Formula

♠ **Proposition 1.2.1.1** (Cauchy) If f is a function defined in a neighborhood U of a closed polydisc $\bar{\Delta}(a,r) = \prod D(a_n, r_n)$ and holomorphic in(contains) a neighborhood U and holomorphic in each variable z_i at each point U, then

$$f(z) = (\frac{1}{2\pi i})^n \int_{|\xi_n - a_n| = r_n} \cdots \int_{|\xi_1 - a_1| = r_1} \frac{f(\xi_1, \ldots, \xi_n)}{\prod(\xi_i - z_i)} d\xi_i, z \in \bar{\Delta}(a,r)$$

∘ **Proof** Omitted,from Cauchy's formula in one variable. □

Note that:

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♠ **Proposition 1.2.1.2** (Osgood's Lemma)

∘ **Proof** ... □

♠ **Proposition 1.2.1.3** (Cauchy's Inequality)

1.2.2 Hartog's Theorem

♠ **Proposition 1.2.2.1** (Jensen's Inequality)

∘ **Proof** ... □

♠ **Proposition 1.2.2.2** ...

∘ **Proof** ... □

1.2.3 Convergence Theorem

1.3 Domains of Holomorphy